SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1002	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	21-08-2026	Duration:	60 minutes

Code of Conduct

*I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: R.Pranathi

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)

Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4

Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1. INTRODUCTION

The **Smart Construction Project Management System (SCPMS)** is a software system designed to manage and monitor construction projects efficiently. Construction projects involve multiple activities such as project planning, task assignment, worker management, material management, budget monitoring, scheduling, and progress tracking.

The proposed system provides a centralized platform where project managers, site engineers, supervisors, workers, and administrators can manage construction-related activities. The system also provides notifications, dashboards, reports, and alerts to support timely decision-making.

Testing is required to ensure that the system performs correctly under normal, boundary, invalid, and exceptional conditions. The test case design verifies the **correctness, functionality, reliability, usability, security, and performance** of the proposed system.

2. OBJECTIVES OF TESTING

The main objectives are:

1. To verify that all functional requirements work correctly.
2. To verify that non-functional requirements such as usability, security, reliability, and performance are satisfied.
3. To identify defects before system deployment.
4. To verify valid and invalid user inputs.
5. To test boundary conditions and exceptional situations.
6. To verify role-based access control.
7. To verify project, task, worker, material, budget, and schedule management.
8. To ensure accurate project progress and cost calculations.
9. To verify notifications and report generation.
10. To establish complete traceability between requirements and test cases.

3. SYSTEM OVERVIEW

The Smart Construction Project Management System contains the following major modules:

1. User Authentication
2. Role-Based Access Control
3. Project Management
4. Task Management
5. Worker Management
6. Material Management
7. Budget and Expense Management
8. Schedule Management
9. Progress Tracking
10. Notification and Alert Management
11. Dashboard and Report Generation
12. Search and Filtering
13. Error Handling

Main Users

User	Responsibilities
Administrator	Manage users, roles, and system settings
Project Manager	Create projects, assign tasks, monitor budget and progress
Site Engineer	Monitor construction activities and technical progress
Supervisor	Manage workers and daily tasks
Worker	View assigned tasks and update task status

4. FUNCTIONAL REQUIREMENTS

Requirement ID	Functional Requirement
FR01	The system shall allow registered users to log in using valid credentials.
FR02	The system shall reject invalid login credentials.
FR03	The system shall allow users to log out securely.
FR04	The system shall provide role-based access control.
FR05	The Project Manager shall be able to create a new project.
FR06	The system shall allow project details to be updated.
FR07	The system shall allow authorized users to delete or cancel a project.
FR08	The system shall allow tasks to be created for a project.
FR09	The system shall allow tasks to be assigned to workers.
FR10	The system shall allow task status to be updated.
FR11	The system shall allow worker information to be added and updated.
FR12	The system shall maintain construction material details.
FR13	The system shall track material quantity and availability.
FR14	The system shall prevent material usage exceeding available stock.
FR15	The system shall record project expenses.
FR16	The system shall calculate the remaining project budget.
FR17	The system shall maintain project start and completion dates.
FR18	The system shall validate project schedules.
FR19	The system shall track project completion percentage.
FR20	The system shall generate deadline and resource alerts.
FR21	The system shall generate project reports.
FR22	The system shall provide project search and filtering.
FR23	The system shall validate mandatory and invalid input fields.
FR24	The system shall store and retrieve project data correctly.

5. NON-FUNCTIONAL REQUIREMENTS

Requirement ID	Non-Functional Requirement
NFR01	The system should provide acceptable response time during normal usage.
NFR02	The system should support multiple users simultaneously.
NFR03	The system should prevent unauthorized access to restricted information.
NFR04	The system should provide a simple and user-friendly interface.
NFR05	The system should maintain data accuracy and consistency.
NFR06	The system should remain stable during normal and exceptional operations.
NFR07	The system should provide clear error and confirmation messages.
NFR08	The system should work correctly on supported browsers and devices.
NFR09	The system should recover correctly after temporary failures.
NFR10	The system should protect sensitive project and user information.

6. TEST SCENARIOS

Scenario ID	Test Scenario
TS01	Verify valid user login.
TS02	Verify invalid login handling.
TS03	Verify logout functionality.
TS04	Verify role-based access control.
TS05	Verify creation of a new construction project.
TS06	Verify modification of project information.
TS07	Verify project deletion/cancellation.
TS08	Verify task creation.
TS09	Verify worker assignment to tasks.
TS10	Verify task status transitions.
TS11	Verify worker registration and modification.
TS12	Verify construction material management.
TS13	Verify material stock availability.
TS14	Verify prevention of excessive material usage.
TS15	Verify budget and expense management.
TS16	Verify remaining budget calculation.
TS17	Verify project schedule creation.
TS18	Verify invalid project schedules.
TS19	Verify project progress tracking.
TS20	Verify deadline and low-stock notifications.
TS21	Verify project report generation.
TS22	Verify project search and filtering.
TS23	Verify input validation.
TS24	Verify system behavior during exceptional conditions.
TS25	Verify performance, usability, security, reliability, and compatibility.

7. TEST DESIGN TECHNIQUES

Four appropriate test design techniques are applied.

7.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups.

Example: Project Progress

Partition	Input	Expected
Invalid	Less than 0%	Reject
Valid	0%–100%	Accept
Invalid	Greater than 100%	Reject

7.2 Boundary Value Analysis

For project progress, the valid boundary is **0% to 100%**.

Test Data	Boundary	Expected Result
-1%	Just below minimum	Reject
0%	Minimum	Accept
1%	Just above minimum	Accept
99%	Just below maximum	Accept
100%	Maximum	Accept
101%	Just above maximum	Reject

7.3 Decision Table Testing

Task Assignment

Conditions:

- Task exists
- Worker available
- User authorized

Rule	Task Exists	Worker Available	User Authorized	Expected Result
R1	Yes	Yes	Yes	Task assigned
R2	Yes	No	Yes	Assignment rejected
R3	No	Yes	Yes	Task not found
R4	Yes	Yes	No	Access denied

This ensures all important combinations are tested.

7.4 State Transition Testing

A task follows the states:

Created → Assigned → In Progress → Completed

An exceptional state is:

In Progress → Delayed

Current State	Action	Expected State
Created	Assign worker	Assigned
Assigned	Start task	In Progress
In Progress	Complete task	Completed
In Progress	Report delay	Delayed
Completed	Restart task	Rejected/Confirmation Required

8. DETAILED TEST CASES

8.1 Login and Access Control

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC001	Valid login	Open login → enter credentials → click Login	Valid Manager ID/password	Manager dashboard is displayed	Not Executed	Not Executed
TC002	Invalid password	Enter valid ID and wrong password → Login	manager01 / wrong123	Login rejected and error message displayed	Not Executed	Not Executed
TC003	Invalid username	Enter unregistered username	unknown01 / Test@123	Login rejected	Not Executed	Not Executed
TC004	Empty credentials	Leave both fields blank → Login	Blank	Mandatory field messages displayed	Not Executed	Not Executed
TC005	Logout	Login → click Logout	Valid user	Session ends and login page appears	Not Executed	Not Executed
TC006	Unauthorized access	Login as Worker → open Manager module	Worker account	Access denied	Not Executed	Not Executed

8.2 Project Management

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC007	Create project	Projects → Add Project → enter details → Save	Green Residency; Chennai; ₹50,00,000	Project created successfully	Not Executed	Not Executed
TC008	Missing project name	Leave name blank → Save	Name = blank	Project not created; validation displayed	Not Executed	Not Executed
TC009	Zero budget	Enter budget 0 → Save	₹0	Budget rejected	Not Executed	Not Executed
TC010	Negative budget	Enter negative amount → Save	-₹10,000	Negative budget rejected	Not Executed	Not Executed
TC011	Update project	Open project → Edit location → Save	Location = Coimbatore	New location stored correctly	Not Executed	Not Executed
TC012	Duplicate project ID	Create project using existing ID	PRJ001	Duplicate ID rejected	Not Executed	Not Executed
TC013	Delete project	Select project → Delete → Confirm	PRJ001	Project deleted/cancelled according to system rules	Not Executed	Not Executed

8.3 Task Management

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC014	Create task	Project → Add Task → Save	Foundation Work	Task created successfully	Not Executed	Not Executed
TC015	Assign available worker	Select task → Assign Worker	Worker = Ravi; Available	Task assigned successfully	Not Executed	Not Executed
TC016	Assign unavailable worker	Select unavailable worker → Assign	Worker = Ravi; Unavailable	Assignment rejected/warning displayed	Not Executed	Not Executed
TC017	Update task status	Open task → Change status	Assigned → In Progress	Status changes successfully	Not Executed	Not Executed
TC018	Complete task	Change status to Completed	In Progress → Completed	Task becomes Completed	Not Executed	Not Executed
TC019	Invalid task dates	Enter end date before start date	Start 20-Aug; End 18-Aug	Invalid schedule rejected	Not Executed	Not Executed
TC020	Restart completed task	Open completed task → Start again	Completed → In Progress	Action rejected or confirmation requested	Not Executed	Not Executed

8.4 Worker Management

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC021	Add worker	Workers → Add → Save	Arun; W101; Supervisor	Worker added successfully	Not Executed	Not Executed
TC022	Missing worker ID	Leave ID blank → Save	ID = blank	Validation error displayed	Not Executed	Not Executed
TC023	Duplicate worker ID	Add worker using existing ID	W101	Duplicate ID rejected	Not Executed	Not Executed
TC024	Update worker	Edit worker details → Save	Role = Site Engineer	Updated information stored	Not Executed	Not Executed

8.5 Material Management

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC025	Add material	Materials → Add → Save	Cement; 100 bags	Material added	Not Executed	Not Executed
TC026	Zero quantity	Enter quantity 0	0 bags	Invalid quantity rejected	Not Executed	Not Executed
TC027	Negative quantity	Enter -5	-5 bags	Negative quantity rejected	Not Executed	Not Executed
TC028	Update stock	Edit material quantity	100 → 150 bags	Quantity becomes 150	Not Executed	Not Executed
TC029	Consume material	Record usage	Available 100; Used 20	Remaining stock = 80	Not Executed	Not Executed
TC030	Excess material usage	Record usage greater than stock	Available 100; Used 120	Transaction rejected; insufficient stock displayed	Not Executed	Not Executed
TC031	Low-stock alert	Reduce stock below threshold	Stock 5; Minimum 10	Low-stock notification generated	Not Executed	Not Executed

8.6 Budget and Expense Management

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC032	Add expense	Budget → Add Expense → Save	₹50,000	Expense recorded	Not Executed	Not Executed
TC033	Calculate remaining budget	Add expense to project	Budget ₹10,00,000; Expense ₹2,00,000	Remaining = ₹8,00,000	Not Executed	Not Executed
TC034	Expense equals budget	Enter expense equal to	₹10,00,000	Remaining = ₹0; appropriate budget warning displayed	Not Executed	Not Executed

		budget				
TC035	Expense exceeds budget	Enter expense greater than budget	₹11,00,000	System rejects/warns according to business rule	Not Executed	Not Executed
TC036	Negative expense	Enter negative expense	-₹5,000	Negative value rejected	Not Executed	Not Executed
TC037	Invalid expense format	Enter letters instead of amount	ABC	Invalid amount message displayed	Not Executed	Not Executed

8.7 Schedule and Progress

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC038	Create schedule	Enter valid start/end dates → Save	20-Aug-2026 to 20-Dec-2026	Schedule saved	Not Executed	Not Executed
TC039	End date before start	Enter invalid dates	Start 20-Dec; End 20-Aug	Schedule rejected	Not Executed	Not Executed
TC040	Progress = 0%	Enter progress	0%	Value accepted	Not Executed	Not Executed
TC041	Progress = 1%	Enter progress	1%	Value accepted	Not Executed	Not Executed
TC042	Progress = 99%	Enter progress	99%	Value accepted	Not Executed	Not Executed
TC043	Progress = 100%	Enter progress	100%	Value accepted and completion status updated when conditions are satisfied	Not Executed	Not Executed
TC044	Progress = -1%	Enter progress	-1%	Value rejected	Not Executed	Not Executed
TC045	Progress = 101%	Enter progress	101%	Value rejected	Not Executed	Not Executed

8.8 Notifications and Reports

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC046	Deadline notification	Create task with approaching deadline	Deadline = Tomorrow	Deadline alert generated	Not Executed	Not Executed
TC047	Low-stock notification	Reduce material below minimum	Cement = 5; Minimum = 10	Low-stock alert generated	Not Executed	Not Executed
TC048	Generate report	Reports → Select project → Generate	Green Residency	Complete project report generated	Not Executed	Not Executed

TC049	Empty project report	Generate report for project with no data	Empty project	Empty report/message displayed without crash	Not Executed	Not Executed
TC050	Search project	Enter project name	Green Residency	Matching project displayed	Not Executed	Not Executed
TC051	Search nonexistent project	Enter unknown name	XYZ Project	"No results found" displayed	Not Executed	Not Executed
TC052	Filter by status	Select Completed filter	Status = Completed	Only completed projects displayed	Not Executed	Not Executed

9. EXCEPTIONAL CONDITION TEST CASES

ID	Exceptional Condition	Test Steps	Test Data	Expected Result	Actual Result	Status
TC053	Network failure	Disconnect network while saving project	Valid project	Error displayed; system does not crash	Not Executed	Not Executed
TC054	Database unavailable	Open project while database unavailable	PRJ001	Suitable database error displayed	Not Executed	Not Executed
TC055	Session timeout	Remain inactive until session expires	Valid user	User redirected to login	Not Executed	Not Executed
TC056	Simultaneous update	Two users edit same project	PRJ001	Data consistency maintained	Not Executed	Not Executed
TC057	Invalid file upload	Upload unsupported file	.exe file	File rejected	Not Executed	Not Executed
TC058	Server restart	Restart system → open saved project	PRJ001	Saved data remains available	Not Executed	Not Executed

10. NON-FUNCTIONAL TEST CASES

ID	Test Area	Test Steps	Test Data	Expected Result	Actual Result	Status
NTC001	Performance	Login and load dashboard	Normal user	Dashboard loads within defined response-time requirement	Not Executed	Not Executed
NTC002	Performance	Perform repeated searches	100 searches	System remains responsive	Not Executed	Not Executed
NTC003	Concurrent users	Login using multiple user accounts simultaneously	Multiple users	System supports configured number of users without failure	Not Executed	Not Executed

NTC004	Security	Worker attempts Manager URL	Worker account	Access denied	Not Executed	Not Executed
NTC005	Security	Enter malicious/invalid input	Script-like input	Input is safely handled and system remains secure	Not Executed	Not Executed
NTC006	Usability	New user creates project	Normal project data	User can complete operation with minimal assistance	Not Executed	Not Executed
NTC007	Compatibility	Open system on supported browsers	Chrome/Edge/Firefox	System functions correctly	Not Executed	Not Executed
NTC008	Reliability	Save and reopen project	PRJ001	Data remains accurate	Not Executed	Not Executed
NTC009	Error Handling	Disconnect network during submission	Valid project	Clear error message displayed	Not Executed	Not Executed
NTC010	Recovery	Restart application after failure	Existing project	System recovers and saved data remains available	Not Executed	Not Executed

11. EQUIVALENCE PARTITIONING TEST CASES

11.1 Project Budget

Assume the system accepts budget values greater than ₹0.

Test Case	Input Class	Test Data	Expected Result
EP01	Invalid	-₹1,000	Reject
EP02	Invalid	₹0	Reject
EP03	Valid	₹1	Accept
EP04	Valid	₹5,00,000	Accept
EP05	Invalid	ABC	Reject
EP06	Invalid	Blank	Reject

11.2 Project Progress

Test Case	Input Class	Test Data	Expected Result
EP07	Invalid	-10%	Reject
EP08	Valid	50%	Accept
EP09	Valid	90%	Accept
EP10	Invalid	110%	Reject
EP11	Invalid	ABC	Reject
EP12	Invalid	Blank	Reject

12. BOUNDARY VALUE ANALYSIS

Project Progress Boundary

Valid range: **0%–100%**

Test Case	Input	Expected Result
BVA01	-1%	Reject
BVA02	0%	Accept
BVA03	1%	Accept
BVA04	99%	Accept
BVA05	100%	Accept
BVA06	101%	Reject

Material Quantity Boundary

Assume the minimum valid quantity is **1 unit**.

Test Case	Input	Expected Result
BVA07	-1	Reject
BVA08	0	Reject
BVA09	1	Accept
BVA10	2	Accept

13. DECISION TABLE TESTING

Task Assignment

Rule	Task Exists	Worker Available	User Authorized	Expected Result
R1	Yes	Yes	Yes	Assign task
R2	Yes	No	Yes	Reject assignment
R3	No	Yes	Yes	Display task-not-found
R4	Yes	Yes	No	Display access denied

Test Cases

ID	Combination	Expected Result
DT01	Yes + Yes + Yes	Task assigned
DT02	Yes + No + Yes	Assignment rejected
DT03	No + Yes + Yes	Task-not-found message
DT04	Yes + Yes + No	Access denied

14. STATE TRANSITION TESTING

Task Lifecycle

Created → Assigned → In Progress → Completed

Alternative state:

In Progress → Delayed

ID	Current State	Event	Expected State
ST01	Created	Assign worker	Assigned
ST02	Assigned	Start work	In Progress
ST03	In Progress	Complete work	Completed
ST04	In Progress	Report delay	Delayed
ST05	Completed	Start work again	Rejected/Confirmation Required
ST06	Created	Complete without assignment	Rejected

15. REQUIREMENT TRACEABILITY MATRIX

Requirement	Related Scenarios	Test Cases
FR01	TS01	TC001
FR02	TS02	TC002–TC004
FR03	TS03	TC005
FR04	TS04	TC006
FR05	TS05	TC007–TC010
FR06	TS06	TC011
FR07	TS07	TC013
FR08	TS08	TC014
FR09	TS09	TC015–TC016
FR10	TS10	TC017–TC020
FR11	TS11	TC021–TC024
FR12	TS12	TC025–TC028
FR13	TS13	TC029–TC031
FR14	TS14	TC030
FR15	TS15	TC032, TC036–TC037
FR16	TS16	TC033–TC035
FR17	TS17	TC038
FR18	TS18	TC019, TC039
FR19	TS19	TC040–TC045
FR20	TS20	TC046–TC047
FR21	TS21	TC048–TC049
FR22	TS22	TC050–TC052
FR23	TS23	TC008–TC010, TC022–TC023, TC026–TC027, TC035–TC037, TC044–TC045
FR24	TS24	TC053–TC058

16. NON-FUNCTIONAL REQUIREMENT TRACEABILITY

Requirement	Test Cases
NFR01 – Performance	NTC001–NTC002
NFR02 – Multiple Users	NTC003
NFR03 – Security	NTC004–NTC005
NFR04 – Usability	NTC006
NFR05 – Data Consistency	NTC008
NFR06 – Stability	TC053–TC058
NFR07 – Error Messages	TC002–TC004, TC053–TC057
NFR08 – Compatibility	NTC007
NFR09 – Recovery	NTC010
NFR10 – Data Protection	NTC004–NTC005

17. TEST COVERAGE

The test cases provide coverage for:

Coverage Area	Coverage
Functional Requirements	**100% mapped**
Non-Functional Requirements	**100% mapped**
Login & Access Control	Covered
Project Management	Covered
Task Management	Covered
Worker Management	Covered
Material Management	Covered
Budget Management	Covered
Scheduling	Covered
Progress Tracking	Covered
Notifications	Covered
Reports	Covered
Search & Filtering	Covered
Normal Conditions	Covered
Boundary Conditions	Covered
Invalid Conditions	Covered
Exceptional Conditions	Covered
Equivalence Partitioning	Applied
Boundary Value Analysis	Applied
Decision Table Testing	Applied
State Transition Testing	Applied

Test Case Count

- Functional test cases: **58**
- Non-functional test cases: **10**
- Additional technique-specific test cases: **28**
- Total planned test cases: **96**

18. TEST EXECUTION STATUS

Since this document is a **Test Case Design document**, the test cases have been prepared for execution but have not been executed against a running implementation.

Therefore:

Actual Result: Not Executed

Pass/Fail Status: Not Executed

After executing the system, the tester should replace these values with the actual observed result and **Pass** or **Fail**.

19. ENTRY CRITERIA

Testing can begin when:

1. The system is developed and available for testing.
2. Functional requirements are finalized.
3. Test environment is configured.
4. Test database is available.
5. Test user accounts are created.
6. Required test data is prepared.
7. Major system modules are accessible.

20. EXIT CRITERIA

Testing can be completed when:

1. All planned test cases are executed.
2. All critical test cases have passed.
3. Critical and high-severity defects are resolved.
4. Regression testing is completed.
5. All major functional requirements are verified.
6. Test results are documented.
7. No critical unresolved defects remain.

21. DEFECT SEVERITY CLASSIFICATION

Severity	Description	Example
Critical	Complete system or major module unavailable	System cannot log in any user
High	Major functionality fails	Project cannot be created
Medium	Important function partially fails	Report contains incorrect calculation
Low	Minor issue with limited impact	Alignment or spelling issue

22. SAMPLE TEST EXECUTION FORMAT

After actual execution, the result can be recorded as follows:

Test Case ID	Expected Result	Actual Result	Status
TC001	Dashboard displayed	Dashboard displayed successfully	Pass
TC009	Zero budget rejected	Error message displayed	Pass
TC030	Excess usage rejected	System rejected transaction	Pass
TC033	Remaining = ₹8,00,000	₹8,00,000 displayed	Pass
TC045	101% rejected	System displayed invalid percentage	Pass

CONCLUSION

The **Smart Construction Project Management System** requires comprehensive testing because it manages important construction activities including projects, workers, tasks, materials, budgets, schedules, and progress.

This test case design covers all major functional and non-functional requirements of the proposed system. Normal, boundary, invalid, and exceptional conditions have been considered to ensure comprehensive testing.

The document applies Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing to systematically identify effective test conditions. The Requirement Traceability Matrix connects requirements with test scenarios and test cases, providing strong requirement coverage.

The planned test cases therefore provide a structured approach to verify the correctness, functionality, reliability, usability, security, performance, and data consistency of the Smart Construction Project Management System.

Overall, the test design aims to achieve complete requirement coverage and identify defects before deployment of the system.